

7. Integration Techniques

7.4. Integration by Partial Fractions

7.4a Learning Objectives

- I can understand how to rewrite rational functions as a sum of partial fractions.
- I can understand when and how to integrate by partial fractions.

Theorem 7.4.1 Fundamental Theorem of Algebra

If $P(x)$ is a polynomial with real coefficients, then P can be factored into linear $(ax + b)$ and irreducible quadratic $(ax^2 + bx + c)$ factors.

$$P(x) = \underbrace{(a_1x + b_1) \dots (a_nx + b_n)}_{\text{linear}} \underbrace{(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)}_{\text{quadratic}}$$

7.4a Section Summary: